1. (30 points) A ride-share company charges a fixed "booking fee" plus a constant rate per mile.

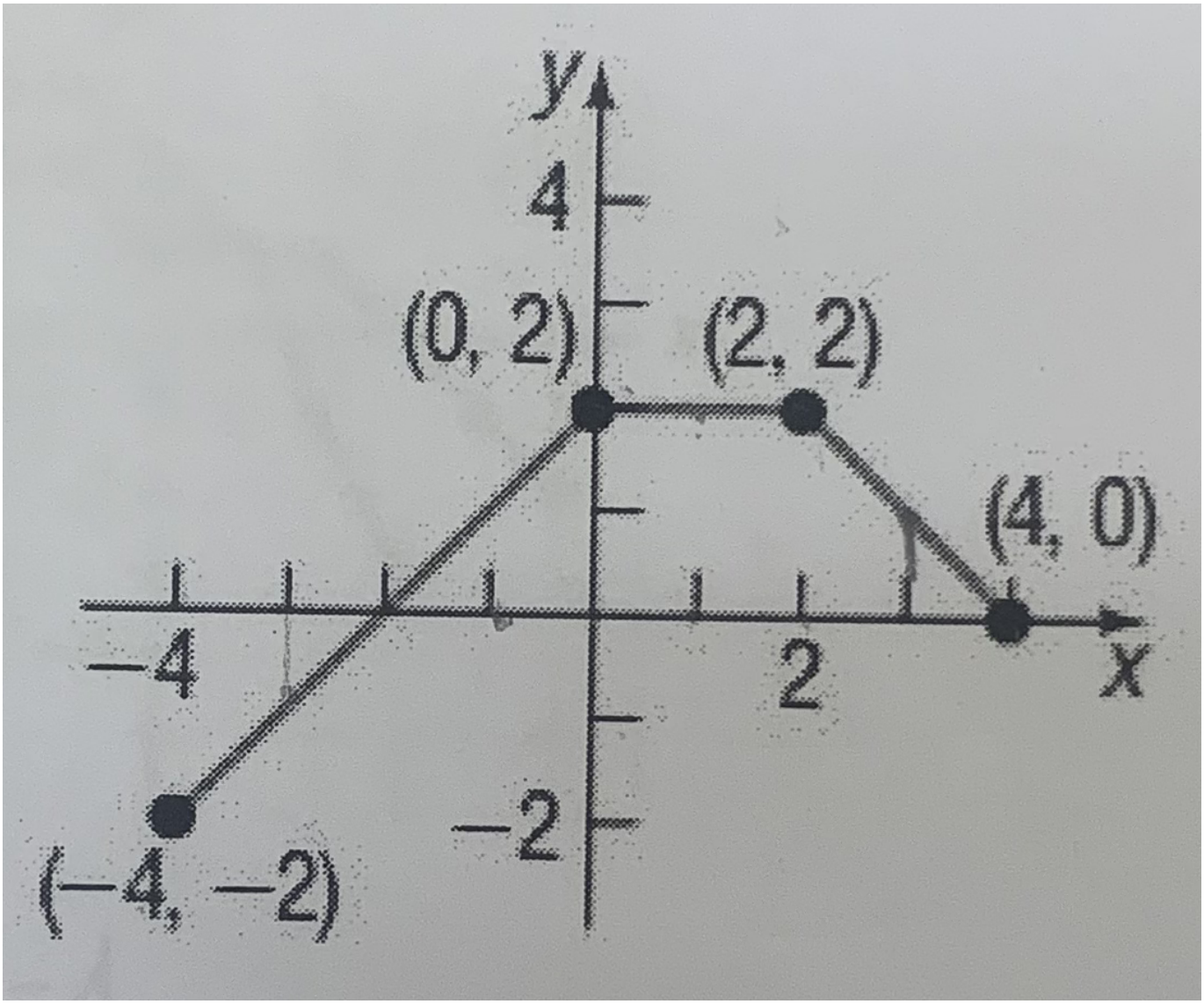
- A 6-mile trip costs $19.80.
- A 14-mile trip costs $39.40.

(a) Find a linear function $C(d)$ that gives the cost (in dollars) as a function of distance d (in miles).
(b) Interpret the slope and y-intercept of your function in context.
(c) Use your model to find the cost of a 25-mile trip.

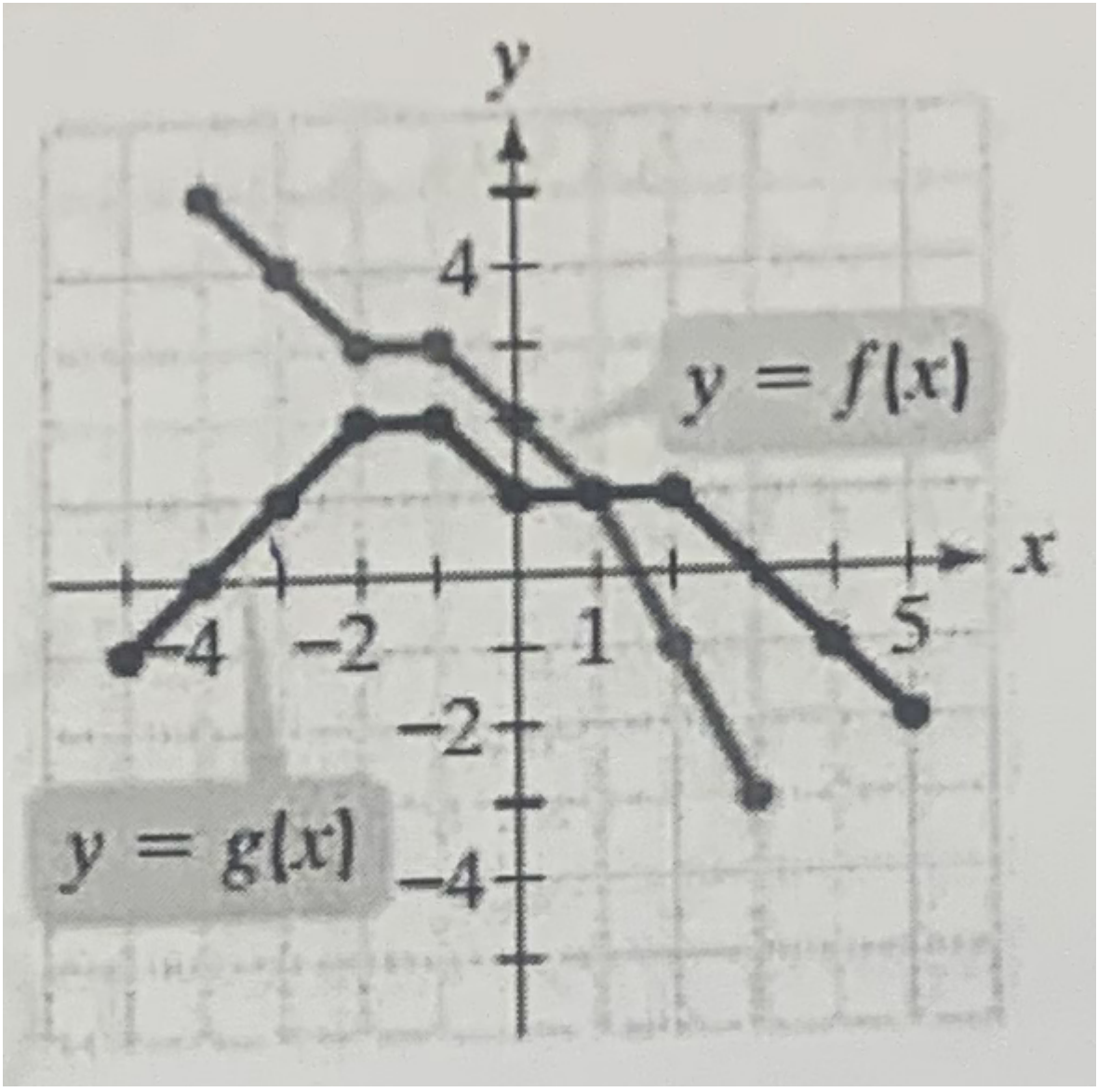
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2. (30 points) Below is the graph of the function $y = f(x)$.
Draw the graph of the following functions:

a. $y = f(x - 2) + 3$
b. $y = f(1 - x)$



3. (30 points) Use the graph of $f(x)$ and $g(x)$ in the picture below to find

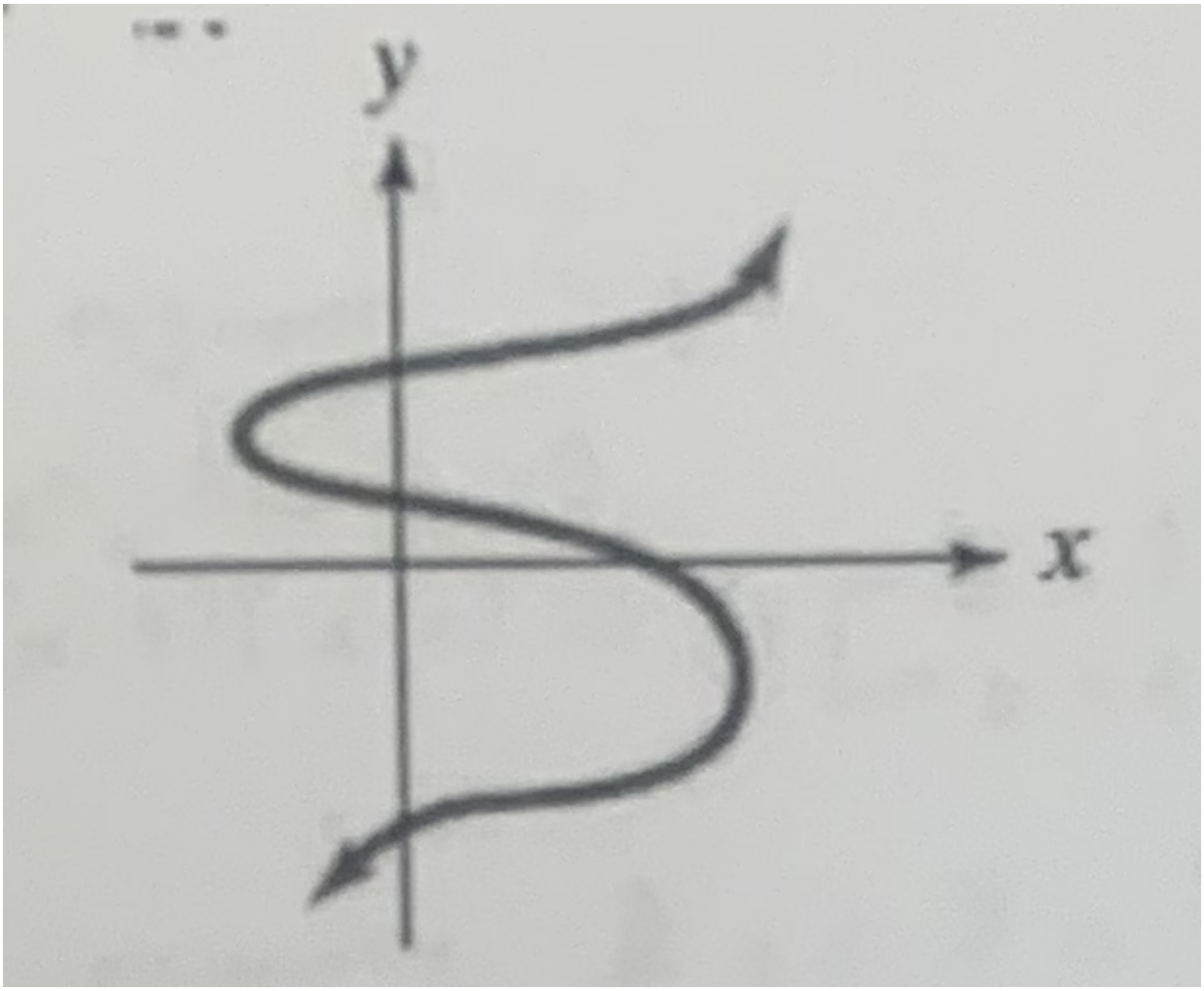


a. $(g - f)(-3)$
b. $(fg)(2)$
c. $(f \circ g)(-2)$
d. $(g \circ f)(3)$
e. $(g/f)(2)$

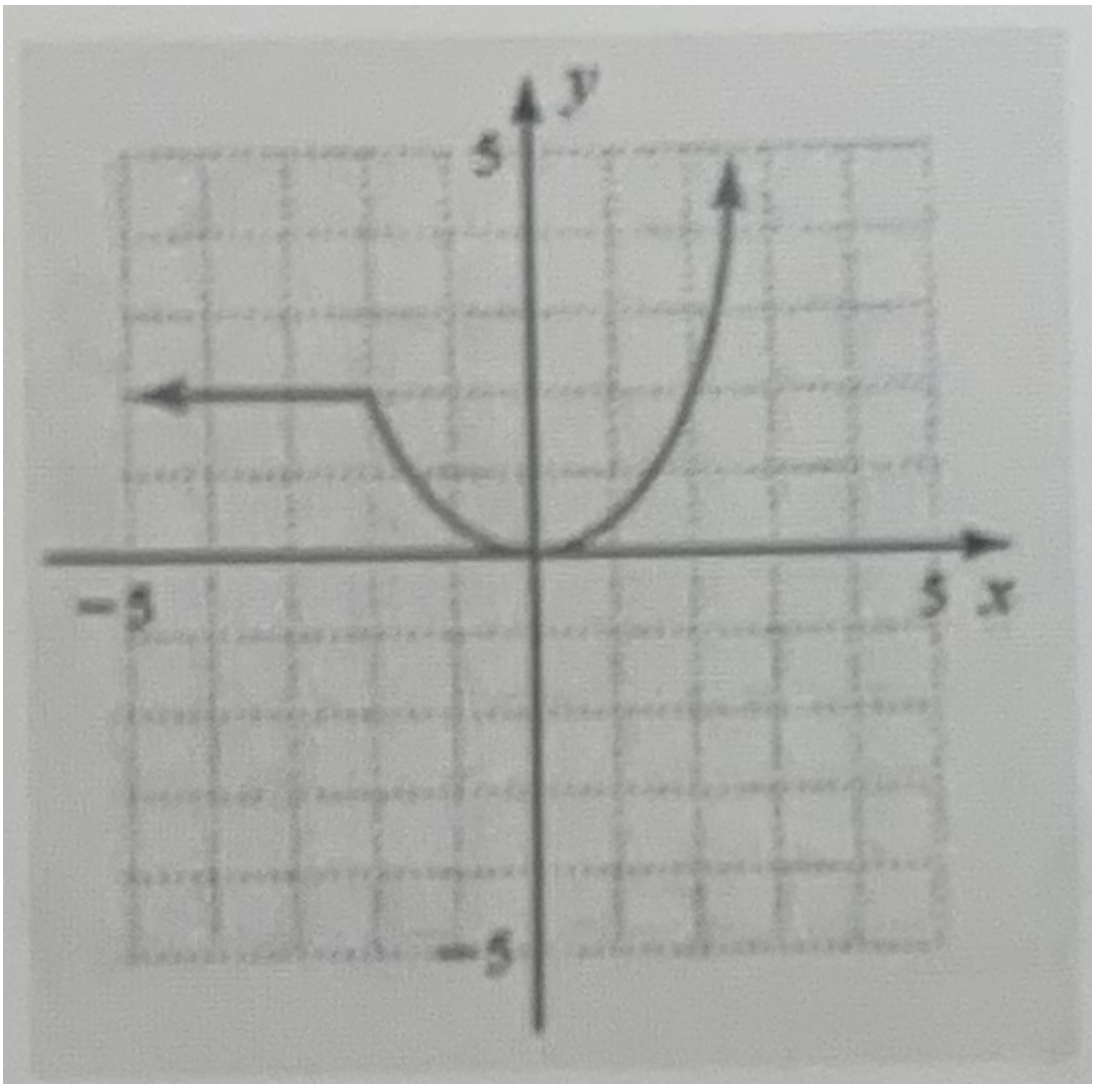
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4. (30 points) Determine if the following equations and graphs represents y as a function of x? Explain your answer.

a. $xy + 2y - 1 = 0$
b.



c.



5. (40 points)

a) Find the asymptotes of the following functions $\frac{2x^2-3x+5}{x^2-4x+3}$

b) Use the synthetic division to divide:
$\frac{2x^4-x^3+5x^2-2}{x+1}$

6. (30 points) You want to accumulate money for a graduate school fund. The account earns 5.2% interest compounded continuously.

- You deposit an unknown amount P today ($t = 0$).
- After 4 years, you deposit an additional $3,000 into the same account.
- After a total of 10 years from your first deposit, you want the account to contain exactly $20,000.

Assume the account compounds continuously at rate $r = 0.052$.

(a) Write an expression for the account balance $A(10)$ at $t = 10$ years in terms of P, and set it equal to 20,000.
(b) Solve your equation to find the required initial deposit P. You may leave your answer in exact form involving exponentials and natural logarithms.

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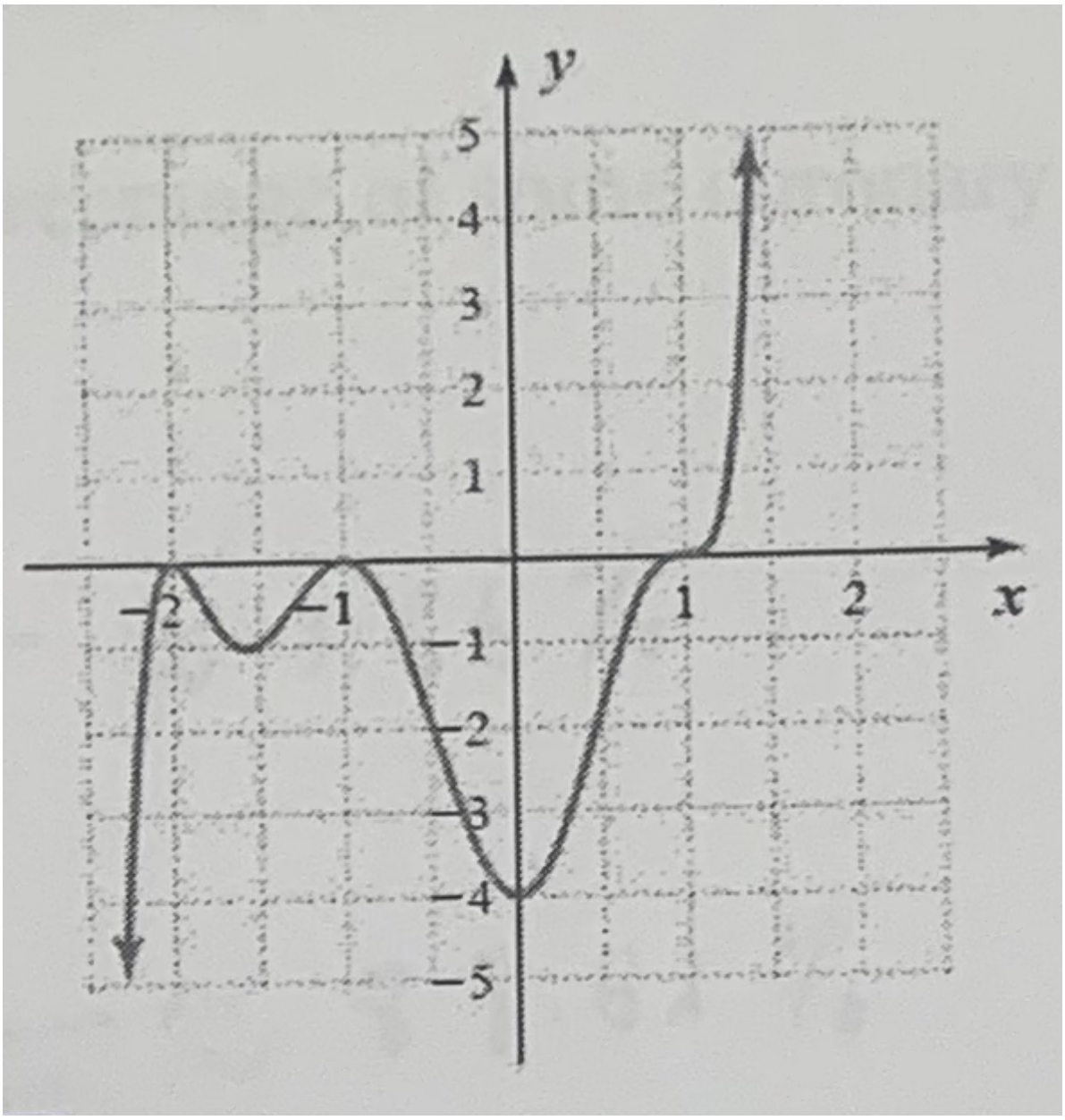
7. (40 points)

a) Solve the following inequality and draw the solution sets on the real line: $\frac{2x}{x+3} \leq \frac{1}{x}$

b) Solve the equation $\ln(x+1) - \ln(2x-4) = \ln 4$

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8. (40 points) For the polynomial graph below (Assume all zeroes are real).



(a) state whether the degree of the function is even or odd;
(b) use the graph to name the zeroes of f, then state whether their multiplicity is even or odd;
(c) state the minimum possible degree of f and write one possible function for f in factored form;
(d) estimate the domain and range.

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9. (30 points) The logistic growth function $P(x) = \frac{90}{1+271e^{-0.122x}}$

models the percentage, $P(x)$, of Americans who are x years old with some coronary heart disease.

a) What percentage of 20-year-olds have some coronary heart disease?
b) What percentage of 80-year-olds have some coronary heart disease?

c) At what age is the percentage of some coronary heart disease 50%?